

Shubham Amrawat

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PROFESSIONAL SUMMARY

Highly motivated **Associate Software Developer** specializing in the **MERN/MEAN stack**, React Native, and advanced AI integration (OpenAI, Gemini APIs). Proven ability to deliver high-impact, full-stack applications, evidenced by work in Punjab Government-backed entrepreneurship platform. Expertise in building robust authentication flows, scalable services, and real-time features. Eager to leverage strong technical skills, communication, and debugging ability to solve complex problems and drive product success.

TECHNICAL SKILLS

Languages: Java, C++, JavaScript, HTML5, CSS3, TypeScript

Frontend/UI: React.js, Angular.js, React Native, Tailwind CSS, DaisyUI

Backend/Runtime: Node.js, Express.js

Database/Cloud: MongoDB, Cloudinary

AI/APIs: OpenAI API (GPT-4o), Google Gemini API, Claude API, FCM Messaging

Tools & Version Control: Git, GitHub, VS Code, Vercel, Render, Postman API

PROFESSIONAL EXPERIENCE

Venturepact Outgrow Private Limited

Associate Software Developer (MERN/MEAN Stack, React Native, AI Integration)

New Delhi, India

Nov 2024 – Present

- **Full-Stack Development:** Delivering features using the **MERN/MEAN stack**, **React Native**, and integrating advanced AI capabilities via **OpenAI** and **Gemini APIs**.
- **Code Quality & Optimization:** Contributed to codebase optimization by fixing a variety of bugs, collaborating on new features, and **optimizing numerous APIs** for quicker response times.

Key Projects:

PUNJAB Entrepreneurship Mindset Program (*Web & Android App - Live Production*):

- Engineered secure, end-to-end **authentication and onboarding flows** for both the native Android and Web applications.
- Developed the critical **quiz submission flow** across platforms, essential for evaluating student scores and course progress.
- Implemented the **Helpdesk chat functionality** in the Android app, allowing students to raise queries and interact with admins for resolution.
- Built and deployed reliable **FCM Messaging**-based notification functionality in the Android application.
- Designed and implemented a **Milestone and Term Lock Service** architecture to regulate and control student course access, ensuring structured progress.
- Built the profile management flow, including edit functionality and change password service, and implemented a **revenue tracking service**.

MUAI V2 - Masters Unions Educational Platform (*MEAN Stack*):

- Established a complete **authentication setup**, including role-based redirection, login, signup, and link-based forgot password.
- Implemented an **AI-driven "Word of the Day" service** by leveraging the **Gemini API** to fetch fresh, relevant MBA content (word, meaning, usage, pronunciation).
- Designed and built an **Achievements and Badges service** (backend and frontend) that automatically awards badges based on specific user logic and checks.
- Developed the **Batchmate Service**, including a **GPA-based leaderboard** showing the top 10 students and real-time online status.
- Built comprehensive profile-related **CRUD** operations service and corresponding frontend implementations.

PERSONAL PROJECTS

SKTCH.AI (sktchai.com): *AI-powered UI builder: draw a wireframe, upload a sketch, or describe your idea — get production-ready React + Tailwind components instantly.*

- **Tech Stack:** React 19, TypeScript, Vite, Tailwind CSS v4; Node.js/Express, MongoDB Atlas, JWT (rotating refresh tokens); **OpenAI GPT-4o**, **Anthropic Claude**, **Google Gemini 2.5 Flash**; Cloudflare R2, Cloudinary; Deployed on Vercel + Render.
- Engineered a **multi-model AI generation pipeline** with a two-step chain: cheap GPT-4o Mini pre-analyzes sketches into a structured JSON blueprint, then the selected model (GPT-4o / Claude / Gemini) generates production-ready JSX — reducing prompt token cost by over 60%.
- Built **real-time streaming code generation** via Server-Sent Events (SSE) with custom compression bypass, multi-model routing (claude-* / gemini-* / OpenAI prefixes), and a live split-pane code preview that streams tokens as they arrive.
- Implemented a full **public sketch ecosystem**: Excalidraw-based canvas, R2/S3 snapshot persistence, public gallery with likes/forks/views, explore page with search and trending sort, and shareable sketch URLs.
- Built a **curated Component Library** feature with 28+ plug-and-play React components across 15 categories, a dedicated AI edit endpoint (Gemini 2.5 Flash with surgical-edit system prompt), and one-click **.tsx** export.
- Secured the platform with **JWT access + rotating refresh tokens**, Google OAuth 2.0, OTP-based password reset via SMTP, per-route auth middleware, and API rate limiting (20 AI generations/hr, 200 general requests/15 min).

AURA — Premium Wallpaper Discovery Platform (aurora-walls.com): *Production-grade full-stack wallpaper platform with AI-powered visual search, semantic mood search, and creator ecosystem.*

- **Tech Stack:** Next.js 16 (App Router, RSC), Hono, Drizzle ORM, PostgreSQL, Cloudflare R2, pgvector; Deployed on Vercel + Render behind Cloudflare CDN.
- Architected a **monorepo** (pnpm workspaces + Turborepo) with shared **@aura/types**, **@aura/db** packages, enabling type-safe full-stack development across web and API services.
- Built a custom **OTP-based authentication system** from scratch (email → OTP via Brevo SMTP → JWT → httpOnly cookies), deliberately chosen over managed auth for learning depth.
- Implemented **AI-powered visual similarity search** using **CLIP embeddings** (Xenova/clip-vit-base-patch32) stored in **pgvector**, with hybrid scoring (75% CLIP cosine similarity + 15% category + 10% tag overlap).
- Built **semantic and mood-based search** using CLIP text embeddings, enabling natural language queries like "feels like 3am Tokyo" to return visually relevant results.
- Designed a **trending algorithm** using time-decay scoring — $\text{score} = \frac{(\text{downloads} \times 3) + (\text{likes} \times 2) + (\text{views} \times 0.1)}{\text{hours}^{0.8}}$ — pre-computed hourly via background scheduler.
- Engineered a **creator upload pipeline** with parallel AI content moderation (Sightengine), server-side metadata extraction (sharp + blurhash), and automated CLIP embedding generation.

EDUCATION

Netaji Subhas University of Technology (NSUT)

B. Tech, Computer Science with Specialization in Artificial Intelligence

New Delhi, India

2023

Army Public School

12th Grade

New Delhi, India

2019

CERTIFICATIONS

Web Development: Internshala Trainings (93%)

C++ Foundation with Data Structures: Coding Ninjas

Standard Java — MERN Stack: Coding Ninjas